ROOT CLASSES - MATHEMATICS

Class 10th NCERT Study Material & Formulas

Chapter 2: Polynomials (Bahupad)

Key Concepts & Important Formulas:

* Degree of Polynomial: Highest exponent of variable x in p(x). Linear (degree 1), Quadratic (degree 2), Cubic (degree 3).

* Geometric Meaning of Zeroes: Number of zeroes of y = p(x) equals number of points where graph intersects x-axis.

* Relationship between Zeroes and Coefficients of Quadratic Polynomial ax^2 + bx + c:
- Sum of zeroes (alpha + beta) = -b/a
- Product of zeroes (alpha * beta) = c/a

* Forming Quadratic Polynomial: p(x) = k[x^2 - (alpha + beta)x + (alpha * beta)].

Solved Exam Examples & Practice:

Example 1: Find zeroes of x^2 + 7x + 10 and verify relationship with coefficients.
(x + 2)(x + 5) = 0 => zeroes are -2 and -5.
Sum = -2 + (-5) = -7 = -(7)/1 = -b/a.
Product = (-2)*(-5) = 10 = 10/1 = c/a.